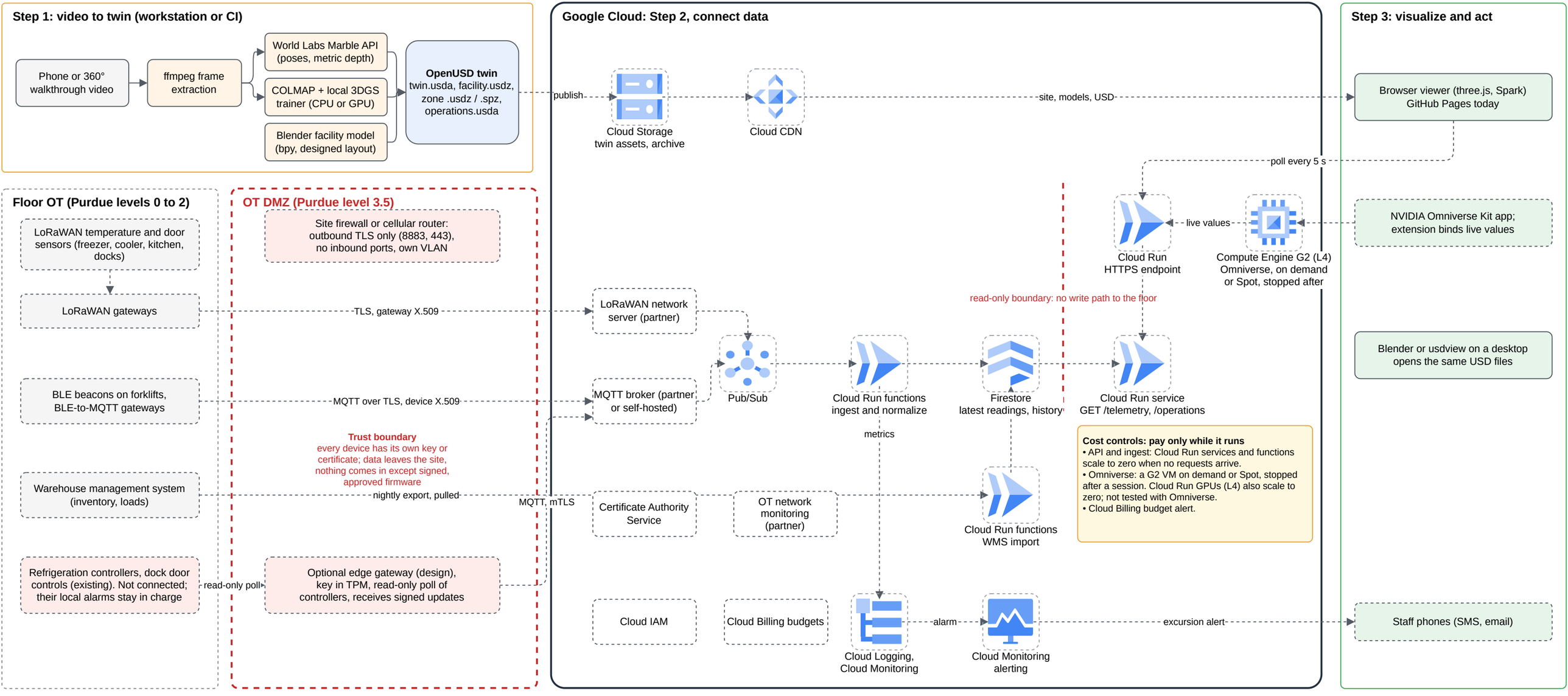
Food bank digital twin on Google Cloud



OT security: zones, conduits and what is built (docs/OT_SECURITY.md)
Field devices and edge gateways sit on the floor; the OT DMZ passes only outbound TLS; cloud ingest accepts data and sends no commands; the twin is read-only. Existing refrigeration and dock controls are not connected. Red dashed lines are trust boundaries. The only designed path toward the floor is a signed, human-approved firmware update.
Built: nothing on Google Cloud; this is a design only.
Design only (dashed): per-device certificates and keys, least-privilege device policies, segmentation, the edge gateway, device monitoring, plausibility and heartbeat checks on readings, signed updates.

Equivalent design, not deployed. Google Cloud IoT Core was retired in August 2023, so devices reach Pub/Sub through a partner or self-hosted MQTT broker and network server.